

Topic: Agroecology

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May 25, 2026

Introduction/Lit Review

1. Rationale for sustainable intensification

Barnosky et al., 2012; Dodds et al., 2013; Eigenbrod et al., 2011; Godfray et al., 2010; Lobell et al., 2011; Parry et al., 2004; Rogers, 2008; Schmidhuber and Tubiello, 2007; Tilman et al., 2001; Wada et al., 2010; Zhao and Running, 2010

(a) Severe constraints

- i. Affluent population demands more meat and dairy
- ii. General public demands reduction in externalities
- iii. Increased weather and climate variability
- iv. Economic efficiency

2. What is sustainable intensification?

Altieri et al., 2015; Baulcombe et al., 2009; Gliessman, 2015; Nicholls and Altieri, 2013

(a) Sustainable yields

(b) Increased soil fertility

(c) Increased soil organic carbon

(d) Increased aboveground and belowground biodiversity

(e) Increased transpiration efficiency

(f) Decreasing negative environmental impacts

3. Need for diagnostic tools for ecological assessment

Compant, Stephane et al., 2010; Prosser et al., 2007; Rygiewicz et al., 2010; Scheller et al., 2007

(a) Evaluation of different land management strategies and across a variety of landscape types

(b) Simulation modeling required to test scenarios

4. Main objectives

Gliessman, 2015; Scheller et al., 2007

(a) build an extension, *AgroEV* ('Agricultural-Ecological Evaluation'), for use with the landscape change model, LANDIS-II

(b) Investigate the conditions that will create the emergent, whole-system properties of sustainable agriculture

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